

Statistical Inference and Modeling

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Semester II, 2025

- Lecture 1: Introduction
- Lecture 2: Model selection
- Lecture 3: Resampling
- Lecture 4: Linear regression
- Lecture 5: Robust regression
- Lecture 6: Classification
- Lecture 7: Logistic regression
- Lecture 8: Neural networks
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- Lecture 10: Bayesian decision, LDA, QDA, and naive Bayes
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- Lecture 13: Bagging, random forest, boosting
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18. Density-based spatial clustering of applications with noise (DBSCAN)

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Problems with K-mean (recap)

- 4 DBSCAN
 - What is DBSCAN?
 - How DBSCAN work?

Supervised vs unsupervised learning

Supervised learning

$$\{X_j, y_j\}_{j=1}^m$$

Using $X_j \rightarrow \hat{y}_j$

Unsupervised learning

$$\{X_j\}_{j=1}^m$$

Using $X_j \rightarrow$ the data behavior

Examples of unsupervised learning

- ① data clustering
- ② dimensionality reduction:
 - data visualization
 - data compression
- ③ data association (rule-based machine learning)

Characteristics of unsupervised learning

- ① No ground truth $\rightarrow$ hard to assess the result from the model.
- ② More subjective $\rightarrow$ no simple goal for the analysis $\rightarrow$ usually based on rules.

Unsupervised learning algorithms

- ① k-mean clustering
- ② hierarchical clustering
- ③ Gaussian Mixture Model (GMM)
- ④ $\Rightarrow$ Density-based spatial clustering of applications with noise (DBSCAN)

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K-mean clustering algorithms (recap)

Target:

$$\underset{C_1, C_2, \dots, C_K}{\operatorname{argmin}} \sum_{k=1}^K \frac{1}{|C_k|} \sum_{j \in C_k} \|\mathbf{x}_j - \bar{\mathbf{x}}_k\|_2^2$$

Algorithm

• Initialization:

- Assume the number of clusters (K).
- randomly assign a number of 1 to K to each training sample.

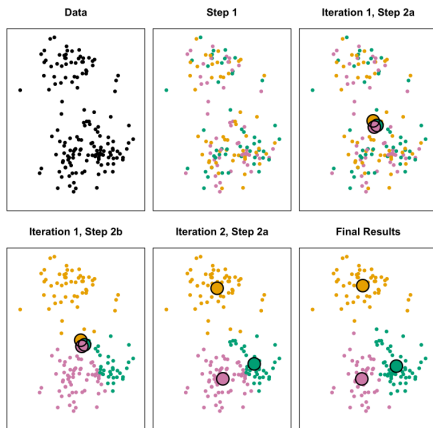
• Start: iterate the following steps until the cluster assignments stop changing

- 1 For each cluster, compute the cluster centroid $\bar{\mathbf{x}}_k$.

$$\bar{\mathbf{x}}_k = \frac{1}{|C_k|} \sum_{j' \in C_k} \mathbf{x}_{j'}$$

- 2 Assign each training sample to the cluster whose centroid is the closest to the training sample.

Visualization



Problems with K-mean

- They make **a hard decision** to which cluster data points belong to ...
- K-mean often provides the data clusters in circular shape, thus, indicating **the lack of the judgement for data density** due to relying solely on the Euclidean distance of the data datapoints.
- Require that the **number of clusters** is predefined. ไม่เหมือน dendrogram.
- Cannot handle **noise** in training samples. noise ในที่นี้คือพวก Outlier, Leverage.

GMM's solution: → ทบทวน Base on Distribution ไม่ใช่ Rule Base แบบ K-Means. ชนิด "Density Base"

- (+) Enabling **soft-decision** by measuring the probability that the data belong to each cluster.
- (+) Enabling **density estimation** by using a **probabilistic model** that can capture the data density of each cluster.
- (0) However, it also cannot handle noise in training samples.
- (0) However, it also requires that the number of clusters is predefined.
- (-) Require **higher computational cost than K-mean**

DBSCAN: คล้าย K-Means ที่เปลี่ยน Rule Base. ชนิด "Density Base"

- (0) Also, make **a hard decision** to which cluster data points belong to ...
- (+) Perform clustering based on the **data density**.
- (+) Can handle noise in training samples. ***
- (+) It do not require that the number of clusters is predefined. ***** } 2 อันนี้ชนะ K-Mean แล้ว.
- (-) Require **more computational cost than K-mean** but not as high as GMM's.

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Problems with K-mean (recap)

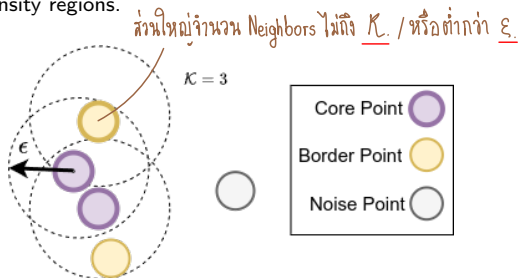
- 4 **DBSCAN**
 - What is DBSCAN?
 - How DBSCAN work?

What is DBSCAN?

⇒ Density-based spatial clustering of applications with noise or DBSCAN determines clusters based on the density of data points.

⇒ DBSCAN puts the data points (input features) into 3 types of points: core points, border points, and noise points:

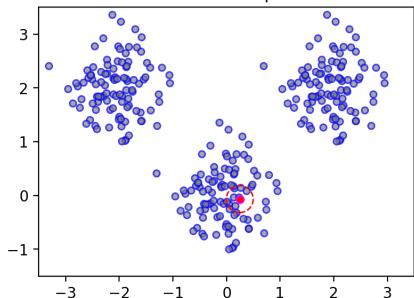
- A **core point** is a point that has at least $\mathcal{K}$ neighbors within a radius ϵ .
- A **border point** falls within the ϵ radius of the core point but is not a core point itself.
- A **noise point** is a point that is neither a core nor a border point. It is often an outlier or a data point in low-density regions.



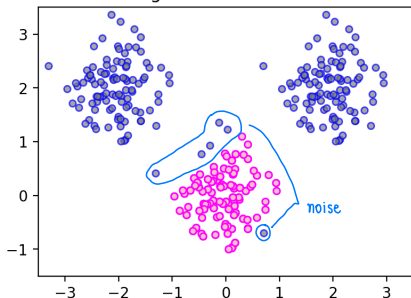
How DBSCAN work?

1. Searching for a new cluster, by finding a core point. A core point is a point that has $\mathcal{K}$ points within ϵ distance of this point: If true, it is a core point. Otherwise, the point is undefined.
2. Once a core point is found, any points within ϵ distance of every point are put together as the same cluster. Thus, a new cluster is established.

Select a core point

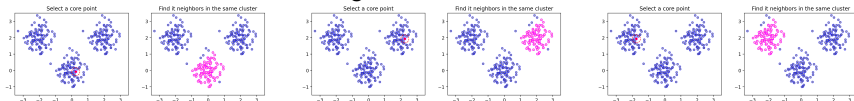


Find its neighbors in the same cluster

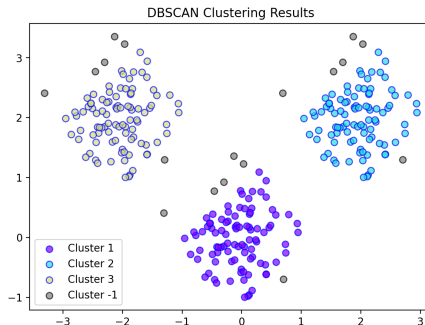


How DBSCAN work?

3. Repeat step 1-2 until all the points in each cluster are labelled. However, any points that have been labeled to a cluster will be ignored in the next round.



4. Points that don't belong to any cluster are labeled as noise.



Coding implementation: [xbeat](#)

K-mean, Hierachical clustering, DBSCAN, GMM

	K-mean	Hierachical clustering	DBSCAN	GMM
Number of clusters?	Yes	No	No	Yes
Hard/Soft decision	Hard	Hard	Hard	Soft
Density/Distance	Distance-based	Distance-based	Density-based	Both
Noise aware	No	No	Yes	No
Complexity	Low	Medium	Medium	High
Hyper-parameters	...	...	...	...
Important assumption (how each method fails)	...	...	...	...

เก็บของ.

คำนวณ distance
ระหว่างทุก Point กับ Centroid

อนุญาตหา #cluster โดย
การตัดสินใจโดยตัวคนเองได้

ทำให้หาค่าได้ #cluster
แบบนี้แหละตามนี้

- [1] Ricardo J. G. B. Campello, Davoud Moulavi, and Joerg Sander. “Density-Based Clustering Based on Hierarchical Density Estimates”. In: *Advances in Knowledge Discovery and Data Mining*. Ed. by Jian Pei et al. Berlin, Heidelberg: Springer Berlin Heidelberg, 2013, pp. 160–172. ISBN: 978-3-642-37456-2.